

Question 1: Identify argument and form.

Consider the argument:

Alice works in Hyderabad during the week (Mon to Fri) and she stays with her parents in Karimnagar on weekends (Sat-Sun).

Therefore, Alice spends her time either in Hyderabad or in Karimnagar.

(a) [1 points] Is this a valid argument or a fallacy? Justify your answer briefly.

(b) [1 points] If it is valid, identify the argument form from the options below; otherwise, choose "None of these".

- (i) Modus Ponens
- (ii) Modus Tollens
- (iii) Hypothetical syllogism
- (iv) Resolution
- (v) None of these

Handwritten notes in Telugu script:

1 - valid argument
2 - simple disjunction
3 - either Hyderabad or Karimnagar

Question 2: Validity of a “wizard–muggle” style argument.

Let the domain be “people”. Use suitable propositional variables.

If a person is a programmer, then they know how to debug code.

If a person is a designer, then they do not debug code.

Y is a designer.

Therefore, Y is not a programmer.

(a) **[1 points]** Translate the above argument into propositional logic using implication(s). (1M)

(b) **[2 points]** Using either a truth table or a proof using rules of inference, determine whether the argument is a valid argument form. Clearly justify. (2M)

Question 3: Tautology check using truth tables.

Let p and q be propositional variables. Consider the formulae:

$$S_1: ((p \rightarrow q) \wedge p) \rightarrow q$$

$$S_2: q \rightarrow (p \rightarrow (p \wedge q))$$

(a) **[2 points]** Using truth tables, determine whether S_1 is a tautology or not. Justify from the table.

(b) **[2 points]** Using truth tables, determine whether S_2 is a tautology or not. Justify from the table. (Show the full truth table(s), but you may organize them neatly to save time.)

Question 4: [4 points] Logical equivalence via truth tables.

The statement $(\neg p) \rightarrow q$ is given. $p \vee q$

Using truth tables, determine which of the following formulae are logically equivalent to $(\neg p) \rightarrow q$. Clearly state "Equivalent" or "Not equivalent" for each, with justification from the table.

I. $p \rightarrow q$ $\sim p \vee q$

II. $q \rightarrow \neg p$ $\sim q \vee \neg p$

III. $(\neg q) \vee \neg p$ $\sim q \vee \neg p$

IV. $(p \wedge \neg q)$ $p \wedge \sim q$

Question 5: Symmetry of Lift for association rules.

You are given an association rule $X \rightarrow Y$ in a transactional database.

(a) **[1 points]** Using the definition of support and confidence based on empirical probabilities, write the formula for $Lift(X \rightarrow Y)$.

(b) **[2 points]** Show algebraically that $Lift(X \rightarrow Y) = Lift(Y \rightarrow X)$. Clearly show each step.

6. Apriori algorithm on a small database.

Consider the following transactional database. Items are represented by integers.

Tx id Items

1 1, 2, 3

2 1, 3, 4

3 2, 3, 4

4 1, 2, 4

5 1, 2, 3, 4

Use the Apriori algorithm with **minimum support = 2 transactions** (absolute count).

- (a) **[3 points]** Show the construction of frequent 1-itemsets, 2-itemsets, and (if any) 3-itemsets using the generate-and-prune steps of Apriori. Clearly show support counts at each level. (3M)
- (b) **[1 points]** List all frequent itemsets returned by Apriori for this database with their support counts. (1M)

2

Question 7: Please answer the following questions:

a) [2+1 points] What is Levinthal's Paradox? How does AlphaFold help to solve this paradox?

if He:

0.5

b) [1+2 points] Name 2 feature selection methods? Please present 3-4 lines description for each of the methods?



AlphaFold

6

- c) **[1.5+1.5 points]** What is the goal of Power transformation ? Please present a diagram with some description that conveys the essence of your explanation ?

10 - 1.5 - 1

- d) **[1+2 points]** What does dimensionality reduction mean ? Please explain how it can be used to reduce overfitting ?

Question 8: Please answer the following questions:

a) [2 points] What is supervised learning? How is it different from unsupervised learning model?

8

b) [2 points] Please provide two methods or algorithms for each of unsupervised and supervised learning models?

c) [4 points] Given that both are learning models, the concept of MSE should possibly be applicable to both – please explain in detail why or why not applicable for each of the models? If MSE is not applicable for any of the two, how would an algorithm/method that falls under the model learn?



Question 9: You are training an ML model based on Bubble sort to perform sorting in ascending order. However, you wrote a ML training program that sorts in descending order. You tested on the following vectors of numbers:

<8, 4, 9, 1, 2>

<4, 3, 7, 2, 9>

<1, 6, 3, 9, 7>

<8, 5, 3, 6, 9>

- a) **[2+2 points]** Assuming your model is accurate for descending order, please compute the Bias and Variance of your program.

b) **[2+2 points]** What would be the Bias and Variance of the model if the actual order we need (i.e., the right order) is Descending order ? Please provide all steps.

c) **[4 points]** As extension to information in (b), Gaussian noise with a mean of 0 and variance of 2 is introduced into the vectors. What would be the MSE of the model if the actual order we need is Descending order ? Please provide all steps.

b) [2+2 points] What would be the Bias and Variance of the model if the actual order we need (i.e., the right order) is Descending order? Please provide all steps.

$$\text{Bias} = 0$$

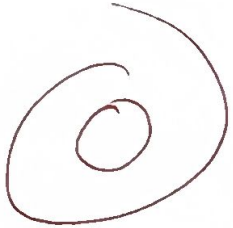
$$\text{Variance} = 0$$

3.5

c) [4 points] As extension to information in (b), Gaussian noise with a mean of 0 and variance of 2 is introduced into the vectors. What would be the MSE of the model if the actual order we need is Descending order? Please provide all steps.

Question 10: We train an ML model for regression. The training and validation Mean Squared Errors (MSE) for every X steps is given below (referred to as stepX where X can be some reasonable number):

StepX	Training MSE	Validation MSE	
1	18.0	20.0	20/80 1.111
2	14.5	15.8	1.089
3	11.0	12.4	1.127
4	8.7	9.8	1.126
5	7.0	8.5	1.1214
6	6.0	8.0	1.33
7	5.3	7.8	1.471
8	4.8	7.7	1.6
9	4.2	7.9	1.8
10	3.9	8.2	2.15
11	3.6	8.6	2.37



Please answer the following questions:

a. **[2 points]** At which stepX is the best model obtained?

b. **[2 points]** Can you compute Bias for the model ? Please provide steps of the computation if yes or the detailed reasoning for why not able to ?

c. **[2 points]** What is the final validation MSE of the selected model along with steps or reasoning involved ?

d. **[2 points]** Please explain why training MSE continues to decrease while validation MSE increases.